Structure from Motion Assignment

Problem 1

The MATLAB function test_essential.m generates a 3D point-cloud in minimum distance of 3m from the camera. A helper function set_rot(alpha,beta,gamma,[dx dy dz]) creates a transformation matrix that is used to move the point-cloud to a coordinate frame of the second camera view. The horizontal u and vertical v coordinates of the projected point-clouds are written alternating into the vectors assuming a uni-focal f/p_x=1 camera.

a) Implement based on the equations in the slides the function essential(u,v) (my version of the solution is provided).

b) Add noise values from in the test_essential.m function to explore the sensitivity of the system to detection accuracy. The noise is added to the calculate ideal projections.

Problem2

The MATLAB function test_homography.m generates a 3D point-cloud in minimum distance of 3m from the camera. A helper function set_rot(alpha,beta,gamma,[dx dy dz]) creates a transformation matrix that is used to move the point-cloud to a coordinate frame of the second camera view. The horizontal u and vertical v coordinates of the projected point-clouds are written alternating into the vectors assuming a uni-focal f/p_x=1 camera.

a) Implement based on equations in the slides the function homography(u,v) (my solution is provided)

b) Add noise values from in the test_homography.m function to explore the sensitivity of the system to detection accuracy. The noise is added to the calculate ideal projections.